# Manage DNS configurations

ONTAP 9.7 REST API reference

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# Manage DNS configurations

## Name-services dns endpoint overview

### Overview

Displays DNS information and controls the DNS subsytem. DNS domain name and DNS servers are required parameters.

### Retrieving DNS information

The DNS GET endpoint retrieves all of the DNS configurations for data SVMs. DNS configuration for the cluster is retrieved via /api/cluster .

### Examples

#### Retrieving all of the fields for all of the DNS configurations

```
# The API:
/api/name-services/dns

# The call:
curl -X GET "https://<mgmt-ip>/api/name-services/dns?fields=*" -H "accept:
application/hal+json"

# The response:
{
"records": [
  {
    "svm": {
      "uuid": "179d3c85-7053-11e8-b9b8-005056b41bd1",
      "name": "vs1"
      "_links": {
        "self": {
          "href": "/api/svm/svms/179d3c85-7053-11e8-b9b8-005056b41bd1"
        }
      }
    },
    "domains": [
      "domainA.example.com"
    ],
    "servers": [
      "10.10.10.10"
    ]
    "_links": {
      "self": {
```

```
        "href": "/api/name-services/dns/179d3c85-7053-11e8-b9b8-
005056b41bd1"
      }
    }
  },
  {
    "svm": {
      "uuid": "19076d35-6e27-11e8-b9b8-005056b41bd1",
      "name": "vs2"
      "_links": {
        "self": {
          "href": "/api/svm/svms/19076d35-6e27-11e8-b9b8-005056b41bd1"
        }
      }
    },
    "domains": [
      "sample.example.com"
    ],
    "servers": [
      "11.11.11.11",
      "22.22.22.22",
      "33.33.33.33"
    ]
    "_links": {
      "self": {
        "href": "/api/name-services/dns/19076d35-6e27-11e8-b9b8-
005056b41bd1"
      }
    }
  }
],
"num_records": 2
"_links": {
  "self": {
    "href": "/api/name-services/dns?fields=*"
  }
}
}
```


**Retrieving all DNS configurations whose domain name starts with *dom**.**

```
# The API:
/api/name-services/dns

# The call:
curl -X GET "https://<mgmt-ip>/api/name-services/dns?domains=dom*" -H
"accept: application/hal+json"

# The response:
{
"records": [
  {
    "svm": {
      "uuid": "179d3c85-7053-11e8-b9b8-005056b41bd1",
      "name": "vs1"
      "_links": {
        "self": {
          "href": "/api/svm/svms/179d3c85-7053-11e8-b9b8-005056b41bd1"
        }
      }
    },
    "domains": [
      "domainA.example.com"
    ]
    "_links": {
      "self": {
        "href": "/api/name-services/dns/179d3c85-7053-11e8-b9b8-
005056b41bd1"
      }
    }
  }
],
"num_records": 1
"_links": {
  "self": {
    "href": "/api/name-services/dns?domains=dom*"
  }
}
}
```

**Retrieving the DNS configuration for a specific SVM**

```
# The API:
/api/name-services/dns/{svm.uuid}

# The call:
curl -X GET "https://<mgmt-ip>/api/name-services/dns/179d3c85-7053-11e8-
b9b8-005056b41bd1" -H "accept: application/hal+json"

# The response:
{
"svm": {
  "uuid": "179d3c85-7053-11e8-b9b8-005056b41bd1",
  "name": "vs1"
  "_links": {
    "self": {
      "href": "/api/svm/svms/179d3c85-7053-11e8-b9b8-005056b41bd1"
    }
  }
},
"domains": [
  "domainA.example.com"
],
"servers": [
  "10.10.10.10"
]
"_links": {
  "self": {
    "href": "/api/name-services/dns/179d3c85-7053-11e8-b9b8-005056b41bd1"
  }
}
}
```

## Creating a DNS configuration

The DNS POST endpoint creates a DNS configuration for the specified SVM.

## Example

The following example shows a POST operation:

```
# The API:
/api/name-services/dns

# The call:
    curl -X POST "https://<mgmt-ip>/api/name-services/dns" -H "accept:
application/hal+json" -H "Content-Type: application/json" -d "{ \"svm\": {
\"uuid\": \"179d3c85-7053-11e8-b9b8-005056b41bd1\" }, \"domains\": [
\"domainA.example.com\" ], \"servers\": [ \"10.10.10.10\" ]}"
```

## Updating a DNS configuration

The DNS PATCH endpoint updates the DNS configuration for the specified SVM.

## Examples

### Updating both the DNS domains and servers

```
# The API:
/api/name-services/dns/{svm.uuid}

# The call:
curl -X PATCH "https://<mgmt-ip>/api/name-services/dns/179d3c85-7053-11e8-
b9b8-005056b41bd1" -H "accept: application/hal+json" -H "Content-Type:
application/json" -d "{ \"domains\": [ \"domainA.example.com\",
\"domainB.example.com\" ], \"servers\": [ \"10.10.10.10\", \"10.10.10.11\"
]}"
```

### Updating the DNS servers only

```
# The API:
/api/name-services/dns/{svm.uuid}

# The call:
curl -X PATCH "https://<mgmt-ip>/api/name-services/dns/179d3c85-7053-11e8-
b9b8-005056b41bd1" -H "accept: application/hal+json" -H "Content-Type:
application/json" -d "{ \"servers\": [ \"10.10.10.10\" ]}"
```

## Deleting a DNS configuration

The DNS DELETE endpoint deletes the DNS configuration for the specified SVM.

## Example

The following example shows a DELETE operation.

```
# The API:
/api/name-services/dns/{svm.uuid}

# The call:
curl -X DELETE "https://<mgmt-ip>/api/name-services/dns/179d3c85-7053-
11e8-b9b8-005056b41bd1" -H "accept: application/hal+json"
```

# Retrieve DNS configurations for all SVMs

GET `/name-services/dns`

Retrieves the DNS configurations of all data SVMs. DNS configuration for the cluster is retrieved and managed via /api/cluster .

## Related ONTAP commands

- `vserver services name-service dns show`
- `vserver services name-service dns check`

## Learn more

- DOC /name-services/dns

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| domains | string | query | False | Filter by domains |
| servers | string | query | False | Filter by servers |
| svm.uuid | string | query | False | Filter by svm.uuid |
| svm.name | string | query | False | Filter by svm.name |
| fields | array[string] | query | False | Specify the fields to return. |
| max_records | integer | query | False | Limit the number of records returned. |
| return_records | boolean | query | False | The default is true for GET calls. When set to false, only the number of records is returned. |

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| return_timeout | integer | query | False | The number of seconds to allow the call to execute before returning. When iterating over a collection, the default is 15 seconds. ONTAP returns earlier if either max records or the end of the collection is reached. |
| order_by | array[string] | query | False | Order results by specified fields and optional [asc |

## Response

```
Status: 200, Ok
```

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| num_records | integer | Number of DNS domain records |
| records | array[dns] | |

**Example response**


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "domains": [
      "example.com",
      "example2.example3.com"
    ],
    "servers": [
      "10.224.65.20",
      "2001:db08:a0b:12f0::1"
    ],
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    }
  }
}
```


## Error

```
Status: Default, Error
```

| Name | Type | Description |
| --- | --- | --- |
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| next | href | |
| self | href | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

svm

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

dns

| Name | Type | Description |
|---|---|---|
| _links | _links | |

| Name | Type | Description |
| --- | --- | --- |
| domains | array[string] | A list of DNS domains. Domain names have the following requirements:<br>• The name must contain only the following characters: A through Z, a through z, 0 through 9, ".", "-" or "_".<br>• The first character of each label, delimited by ".", must be one of the following characters: A through Z or a through z or 0 through 9.<br>• The last character of each label, delimited by ".", must be one of the following characters: A through Z, a through z, or 0 through 9.<br>• The top level domain must contain only the following characters: A through Z, a through z.<br>• The system reserves the following names:"all", "local", and "localhost". |
| servers | array[string] | The list of IP addresses of the DNS servers. Addresses can be either IPv4 or IPv6 addresses. |
| svm | svm | |

error_arguments

| Name | Type | Description |
| --- | --- | --- |
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
| --- | --- | --- |
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |

| Name | Type | Description |
|---|---|---|
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Create DNS domain and server configurations

POST `/name-services/dns`

Creates DNS domain and server configurations for an SVM.

## Important notes

- Each SVM can have only one DNS configuration.
- The domain name and the servers fields cannot be empty.
- IPv6 must be enabled if IPv6 family addresses are specified in the `servers` field.
- Configuring more than one DNS server is recommended to avoid a single point of failure.
- The DNS server specified using the `servers` field is validated during this operation.

The validation fails in the following scenarios:

1. The server is not a DNS server.
2. The server does not exist.
3. The server is unreachable.

## Learn more

- DOC /name-services/dns

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| return_records | boolean | query | False | The default is false. If set to true, the records are returned. |

## Request Body

| Name | Type | Description |
|---|---|---|
| _links | _links | |

| Name | Type | Description |
| --- | --- | --- |
| domains | array[string] | A list of DNS domains. Domain names have the following requirements:<br>• The name must contain only the following characters: A through Z, a through z, 0 through 9, ".", "-" or "_".<br>• The first character of each label, delimited by ".", must be one of the following characters: A through Z or a through z or 0 through 9.<br>• The last character of each label, delimited by ".", must be one of the following characters: A through Z, a through z, or 0 through 9.<br>• The top level domain must contain only the following characters: A through Z, a through z.<br>• The system reserves the following names:"all", "local", and "localhost". |
| servers | array[string] | The list of IP addresses of the DNS servers. Addresses can be either IPv4 or IPv6 addresses. |
| svm | svm | |

#### Example request

```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "domains": [
    "example.com",
    "example2.example3.com"
  ],
  "servers": [
    "10.224.65.20",
    "2001:db08:a0b:12f0::1"
  ],
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  }
}
```

## Response

```
Status: 201, Created
```

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| num_records | integer | Number of DNS domain records |
| records | array[dns] | |

**Example response**


```
{
  "_links": {
    "next": {
      "href": "/api/resourcelink"
    },
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "records": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "domains": [
      "example.com",
      "example2.example3.com"
    ],
    "servers": [
      "10.224.65.20",
      "2001:db08:a0b:12f0::1"
    ],
    "svm": {
      "_links": {
        "self": {
          "href": "/api/resourcelink"
        }
      },
      "name": "svm1",
      "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
    }
  }
}
```


## Error

```
Status: Default
```

ONTAP Error Response Codes

| Error Code | Description |
| --- | --- |
| 2621706 | The specified SVM UUID is incorrect for the specified SVM name |
| 8847360 | Only admin or data SVMs allowed |
| 8847361 | Exceeded the maximum number of domains allowed. Maximum of six domains only |
| 8847362 | Exceeded the maximum number of name servers allowed. Maximum of three name servers only |
| 8847392 | Domain name cannot be an IP address |
| 8847393 | Top level domain name is invalid |
| 8847399 | One or more of the specified DNS servers do not exist or cannot be reached |
| 8847394 | FQDN name violated the limitations |
| 9240587 | FQDN name cannot be empty |
| 9240588 | FQDN name is too long. Maximum supported length: 255 characters |
| 9240590 | FQDN name is reserved. Following names are reserved: "all", "local" and "localhost" |
| 9240607 | One of the FQDN labels is too long. Maximum supported length: 63 characters |
| 13434916 | The SVM is in the process of being created. Wait a few minutes, and then try the command again. |
| 23724130 | Cannot use an IPv6 name server address because there are no IPv6 LIFs |

| Name | Type | Description |
| --- | --- | --- |
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

svm

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

dns

| Name | Type | Description |
|---|---|---|
| _links | _links | |

| Name | Type | Description |
| --- | --- | --- |
| domains | array[string] | A list of DNS domains. Domain names have the following requirements:<br>• The name must contain only the following characters: A through Z, a through z, 0 through 9, ".", "-" or "_".<br>• The first character of each label, delimited by ".", must be one of the following characters: A through Z or a through z or 0 through 9.<br>• The last character of each label, delimited by ".", must be one of the following characters: A through Z, a through z, or 0 through 9.<br>• The top level domain must contain only the following characters: A through Z, a through z.<br>• The system reserves the following names:"all", "local", and "localhost". |
| servers | array[string] | The list of IP addresses of the DNS servers. Addresses can be either IPv4 or IPv6 addresses. |
| svm | svm | |

_links

| Name | Type | Description |
| --- | --- | --- |
| next | href | |
| self | href | |

error_arguments

| Name | Type | Description |
| --- | --- | --- |
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Delete a DNS domain configuration

DELETE `/name-services/dns/{svm.uuid}`

Deletes DNS domain configuration of the specified SVM.

## Related ONTAP commands

- `vserver services name-service dns delete`

## Learn more

- DOC /name-services/dns

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| svm.uuid | string | path | True | UUID of the SVM to which this object belongs. |

## Response

```
Status: 200, Ok
```

## Error

```
Status: Default, Error
```

| Name | Type | Description |
|---|---|---|
| error | error | |

#### Example error

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

### Definitions

#### See Definitions

error_arguments

| Name | Type | Description |
|---|---|---|
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

## Retrieve DNS domain and server configurations

GET `/name-services/dns/{svm.uuid}`

Retrieves DNS domain and server configuration of an SVM. By default, both DNS domains and servers are

displayed. DNS configuration for the cluster is retrieved and managed via /api/cluster .

## Related ONTAP commands

- `vserver services name-service dns show`
- `vserver services name-service dns check`

## Learn more

- DOC /name-services/dns

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| svm.uuid | string | path | True | UUID of the SVM to which this object belongs. |
| fields | array[string] | query | False | Specify the fields to return. |

## Response

```
Status: 200, Ok
```

| Name | Type | Description |
|---|---|---|
| _links | _links | |

| Name | Type | Description |
| --- | --- | --- |
| domains | array[string] | A list of DNS domains. Domain names have the following requirements:<br>• The name must contain only the following characters: A through Z, a through z, 0 through 9, ".", "-" or "_".<br>• The first character of each label, delimited by ".", must be one of the following characters: A through Z or a through z or 0 through 9.<br>• The last character of each label, delimited by ".", must be one of the following characters: A through Z, a through z, or 0 through 9.<br>• The top level domain must contain only the following characters: A through Z, a through z.<br>• The system reserves the following names:"all", "local", and "localhost". |
| servers | array[string] | The list of IP addresses of the DNS servers. Addresses can be either IPv4 or IPv6 addresses. |
| svm | svm | |

#### Example response

```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "domains": [
    "example.com",
    "example2.example3.com"
  ],
  "servers": [
    "10.224.65.20",
    "2001:db08:a0b:12f0::1"
  ],
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  }
}
```

## Error

```
Status: Default, Error
```

| Name | Type | Description |
|---|---|---|
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

svm

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

error_arguments

| Name | Type | Description |
|---|---|---|
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
|---|---|---|
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

# Update DNS domain and server configurations

PATCH `/name-services/dns/{svm.uuid}`

Updates DNS domain and server configurations of an SVM.

## Important notes

- Both DNS domains and servers can be modified.
- The domains and servers fields cannot be empty.
- IPv6 must be enabled if IPv6 family addresses are specified for the `servers` field.
- The DNS server specified using the `servers` field is validated during this operation.

The validation fails in the following scenarios:

1. The server is not a DNS server.
2. The server does not exist.
3. The server is unreachable.

## Learn more

- DOC /name-services/dns

## Parameters

| Name | Type | In | Required | Description |
|---|---|---|---|---|
| svm.uuid | string | path | True | UUID of the SVM to which this object belongs. |

## Request Body

| Name | Type | Description |
|---|---|---|
| _links | _links | |

| Name | Type | Description |
| --- | --- | --- |
| domains | array[string] | A list of DNS domains. Domain names have the following requirements:<br>• The name must contain only the following characters: A through Z, a through z, 0 through 9, ".", "-" or "_".<br>• The first character of each label, delimited by ".", must be one of the following characters: A through Z or a through z or 0 through 9.<br>• The last character of each label, delimited by ".", must be one of the following characters: A through Z, a through z, or 0 through 9.<br>• The top level domain must contain only the following characters: A through Z, a through z.<br>• The system reserves the following names:"all", "local", and "localhost". |
| servers | array[string] | The list of IP addresses of the DNS servers. Addresses can be either IPv4 or IPv6 addresses. |
| svm | svm | |

**Example request**

```
{
  "_links": {
    "self": {
      "href": "/api/resourcelink"
    }
  },
  "domains": [
    "example.com",
    "example2.example3.com"
  ],
  "servers": [
    "10.224.65.20",
    "2001:db08:a0b:12f0::1"
  ],
  "svm": {
    "_links": {
      "self": {
        "href": "/api/resourcelink"
      }
    },
    "name": "svm1",
    "uuid": "02c9e252-41be-11e9-81d5-00a0986138f7"
  }
}
```

## Response

```
Status: 200, Ok
```

## Error

```
Status: Default
```

ONTAP Error Response Codes

| Error Code | Description |
|---|---|
| 8847360 | Only admin or data SVMs allowed |
| 8847361 | Exceeded the maximum number of domains allowed. Maximum of six domains only |

| Error Code | Description |
|---|---|
| 8847362 | Exceeded the maximum number of name servers allowed. Maximum of three name servers only |
| 8847392 | Domain name cannot be an IP address |
| 8847393 | Top level domain name is invalid |
| 8847394 | FQDN name voilated the limitations |
| 8847399 | One or more of the specified DNS servers do not exist or cannot be reached |
| 9240587 | FQDN name cannot be empty |
| 9240588 | FQDN name is too long. Maximum supported length: 255 characters |
| 9240590 | FQDN name is reserved. Following names are reserved: "all", "local" and "localhost" |
| 9240607 | One of the FQDN labels is too long. Maximum supported length: 63 characters |
| 23724130 | Cannot use an IPv6 name server address because there are no IPv6 LIFs |

| Name | Type | Description |
|---|---|---|
| error | error | |

**Example error**

```
{
  "error": {
    "arguments": {
      "code": "string",
      "message": "string"
    },
    "code": "4",
    "message": "entry doesn't exist",
    "target": "uuid"
  }
}
```

## Definitions

**See Definitions**

href

| Name | Type | Description |
|---|---|---|
| href | string | |

_links

| Name | Type | Description |
|---|---|---|
| self | href | |

svm

| Name | Type | Description |
|---|---|---|
| _links | _links | |
| name | string | The name of the SVM. |
| uuid | string | The unique identifier of the SVM. |

dns

| Name | Type | Description |
|---|---|---|
| _links | _links | |

| Name | Type | Description |
| --- | --- | --- |
| domains | array[string] | A list of DNS domains. Domain names have the following requirements:<br>• The name must contain only the following characters: A through Z, a through z, 0 through 9, ".", "-" or "_".<br>• The first character of each label, delimited by ".", must be one of the following characters: A through Z or a through z or 0 through 9.<br>• The last character of each label, delimited by ".", must be one of the following characters: A through Z, a through z, or 0 through 9.<br>• The top level domain must contain only the following characters: A through Z, a through z.<br>• The system reserves the following names:"all", "local", and "localhost". |
| servers | array[string] | The list of IP addresses of the DNS servers. Addresses can be either IPv4 or IPv6 addresses. |
| svm | svm | |

error_arguments

| Name | Type | Description |
| --- | --- | --- |
| code | string | Argument code |
| message | string | Message argument |

error

| Name | Type | Description |
| --- | --- | --- |
| arguments | array[error_arguments] | Message arguments |
| code | string | Error code |

| Name | Type | Description |
| --- | --- | --- |
| message | string | Error message |
| target | string | The target parameter that caused the error. |

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